Finite Fields

A field F that has finitely many elements is called a *finite field*.

There is a natural ring homomorphism $\mathbb{Z} \to F$ (which sends 1 to the multiplicate identity 1 in F). Since the image is a domain, the kernel is a prime ideal in $\mathbb{Z}$. Hence, there is a prime p in $\mathbb{Z}$ such that $\mathbb{Z}/\langle p \rangle$ is a subring of F.

Since $\mathbb{Z}/\langle p \rangle$ is a field and F is a finitely generated module over it, it is a finite dimensional vector space over this field. It follows that there is a positive integer n such that F has $q = p^n$ elements.

> A finite field has $q = p^n$ elements, where n is a positive integer and p is a prime.
>
> A finite field is a vector space over $\mathbb{Z}/\langle p \rangle$.

Multiplicative group

The non-zero elements of F form a commutative group F^* with $q - 1$ elements. It follows that all these elements are roots the polynomial $X^{q-1} - 1$.

A polynomial of degree d over a field can have *at most* d distinct roots. In particular the polynomial $X^d - 1$ has at most d roots in F. If d is the least common multiple of the orders of *all* elements of F^*, then *every* element of F^* satisfies $X^d - 1$. So $d \geq (q - 1)$. Since d divides $q - 1$, we see that $d = q - 1$.

Given two elements a and b of an Abelian group of orders d and e, there is an element whose order is the least common multiple of d and e. Thus, the least common multiple of the orders of all elements of a finite group is the order of *some* element of the group.

Thus, there is an element α of F^* whose order is *exactly* $(q - 1)$.

> The multiplicative group F^* of non-zero elements of a finite field F is a cyclic group.

We see also that every element of F^* is a root of polynomial $X^{q-1} - 1$. It follows that we have:

$$X^q - X = \prod_{\kappa \in F} (X - \kappa) \text{ in } F[X]$$

Thus, the field F with q elements is the splitting field of the polynomial $X^q - X$ over the field $\mathbb{F}_p$. Hence, it is uniquely determined upto isomorphism and we denote it as $\mathbb{F}_q$. In particular, we will use the notation $\mathbb{F}_p = \mathbb{Z}/\langle p \rangle$.

Frobenius endomorphism

Given a commutative ring R with identity that contains $\mathbb{F}_p$ as a subring with the same identity. In this case we say that R is a $\mathbb{F}_p$-*algebra*.

This means that $n \cdot a = 0$ for any a in R when n is an integer divisible by p.

By the Binomial theorem, we see that if a and b are elements of R, then

$$(a+b)^p = \sum_{r=0}^{p} \binom{p}{r} a^r \cdot b^{p-r} = a^p + b^p$$

since $\binom{p}{r}$ is divisible by p for $0 < r < p$.

Now, $(a \cdot b)^p = a^p \cdot b^p$ in any commutative ring R.

> $\phi_p : R \to R$ defined by $\phi(a) = a^p$ is an endomorphism of an $\mathbb{F}_p$-algebra R.

Given a positive integer n, we see that $\phi_q = (\phi_p)^n$ is also an endomorphism. Observe that $\phi_q(a) = a$ if and only if a satisfies the polynomial $X^q - X$.

Note that if $\psi : R \to R$ is an endomorphism of a ring, then the set R^ψ of elements a of R that satisfy $\psi(a) = a$ is a subring.

In particular, the set R^{ϕ_q} is a subring of R.

Minimal polynomial

Given $E \subset F$ fields such that F is a finite dimensional vector space over E; let m be this dimension.

Given an element α of F we obtain a ring homomorphism $E[X] \to F$ which sends X to α. The image $E[\alpha]$ is a finite dimensional vector space of dimension at most m, we see that the kernel of this homomorphism contains a non-zero polynomial of degree at most m. The non-zero polynomial of least degree d that lies in the kernel with leading coefficient 1 is called the *minimal polynomial* of α over E.

Note that, by a simple application of long division, the minimal polynomial divides every other polynomial in the kernel of $E[X] \to E[\alpha]$.

Since $E[\alpha] \subset F$ is a domain, the kernel is generated by an irreducible polynomial of degree d. In other words, the minimal polynomial is an irreducible polynomial of degree d. It follows that $E[\alpha]$ is a field. Clearly, it is the smallest subfield of F that contains E and α.

Existence of an irreducible polynomial of degree n

Given $q = p^n$ a power of a prime p, we have seen that there is a field $\mathbb{F}_q$ which has q elements.

We have seen that there is an element α in $\mathbb{F}_q$ which has multiplicative order $p^n - 1$. Suppose $P(X)$ is the minimal polynomial of α over $\mathbb{F}_p$.

We have seen that $(\mathbb{F}_p)[\alpha]$ is a vector space of dimension $m \leq n$ over $\mathbb{F}_p$; hence, it has p^m elements.

Since $(\mathbb{F}_p)[\alpha]$ is a field, every non-zero element has order dividing $p^m - 1$. In particular, α has order dividing $p^m - 1$. Since α has order $p^n - 1$ and $m \leq n$, it follows that $m = n$.

> There is an irreducible polynomial $P(X)$ of degree n over $\mathbb{F}_p$.
>
> The field $\mathbb{F}_q$ can be identified with $((\mathbb{F}_p)[X])/\langle P(X)\rangle$.

Note that the irreducible polynomial $P(X)$ of degree n need not be *unique*!

Uniqueness upto isomorphism

Since α satisfies $X^q - X$:

> Irreducible polynomials of degree n over $\mathbb{F}_p$ occur among the factors of $X^{p^n} - X$.

If F is *any* field with $q = p^n$ elements, we have seen that $X^q - X$ splits completely in F. It follows that if $Q(X)$ is *any* polynomial which divides $X^{q-1} - 1$, then **q** splits completely in F.

In particular, we see that there is an element β in F which is a root of the irreducible polynomial $Q(X)$ of degree n. As above, we get a homomorphism $\mathbb{F}_p[X] \to F$ which sends X to β and the kernel of this is generated by $Q(X)$.

This gives get a natural injective homomorphism $\mathbb{F}_p[X]/\langle Q(X)\rangle \to F$. Since both sides have the same number of elements, this is an isomorphism.

> Up to isomorphism, there is a *unique* field $\mathbb{F}_q$ with q elements for every prime power q.

Reciprocity

The field $\mathbb{F}_q$ contains a primitive $(q-1)$-th root of unity α for $q = p^n$ a power of a prime p.

Given an integer k which divides $p^n - 1$, we see that $\mathbb{F}_{p^n}$ contains a primitive k-th root of unity (the element $\alpha^{(q-1)/k}$ is such a root of unity).

Thus, the *smallest* n for which $\mathbb{F}_{p^n}$ contains a primitive k-th root of unity is also the smalles n for which k divides $p^n - 1$.

If k divides $p^n - 1$, then k and p are co-prime. Thus, p is an element of the multiplicative group $(\mathbb{Z}/\langle k\rangle)^*$ of invertible elements in $\mathbb{Z}/\langle k\rangle$.

We see that the smallest n as above is the *order* of p as an element of this group.

> The smallest n such that $\mathbb{F}_{p^n}$ contains a primitive k-th root of unity is the order of p as an element of $(\mathbb{Z}/\langle k\rangle)^*$.
>
> For such an n the polynomial $X^k - 1$ splits completely in $\mathbb{F}_{p^n}$.

This is called (*cyclotomic*) *reciprocity* since the problem of solving $X^k - 1$ modulo p is reduced to the behaviour of p in $\mathbb{Z}/\langle k\rangle$.

Its immediate use is that we can work with p quite large as long as k is not too big since we only need to look at $p \bmod k$.

Examples

We look at some examples for small primes p.

The prime $p = 2$. We note that $X^4 - X = X(X-1)(X^2+X+1)$. So the unique irreducible polynomial of degree 2 over $\mathbb{F}_2$ is X^2+X+1. (Alternatively, note that the only other polynomials of degree 2 over $\mathbb{F}_2$ are $X^2+X = X(X+1)$ and $X^2+1 = (X+1)^2$!)

We note that

$$X^8 - X = X(X^7-1) = X(X-1)(X^6+X^5+X^4+X^3+X^2+X+1)$$

which suggests that the last factor is a product of two irreducible polynomials of degree 3 over $\mathbb{F}_2$.

A polynomial of degree 3 over a field F is irreducible if and only if it has no root in F. It follows that X^3+X+1 and X^3+X^2+1 are irreducible polynomials in $\mathbb{F}_2[X]$. This suggests the identity

$$X^6+X^5+X^4+X^3+X^2+X+1 = (X^3+X+1)(X^4+X^2+1) \text{ in } \mathbb{F}_2$$

Observe that

$$X^{16} - X = X(X^15-1) = X\Phi_1(X)\Phi_3(X)\Phi_5(X)\Phi_15(X)$$

We note that $\Phi_3(X) = X^2-X+1$ is of degree 2 so any irreducible polynomial of degree 4 over $\mathbb{F}_2$ can be found as a factor of $\Phi_5(X)\Phi_15(X)$. Since $\mathbb{F}_{16}$ contains a primitive 15-th root of unity, it also contains a primitive 5-th root of unity. This is *not* contained in the sub-field $\mathbb{F}_4$. It follows that $\Phi_5(X) = X^4+X^3+X^2+X+1$ is an irreducible polynomial of degree 4 over $\mathbb{F}_2$. The other two are factors of $\Phi_15(X) \bmod 2$ which is a polynomial of degree $\varphi(15) = 8$.

The prime $p = 3$. We note that $X^9 - X = X(X^8-1) = X(X-1)(X+1)(X^2+1)(X^4+1)$. So an irreducible polynomial of degree 2 over $\mathbb{F}_3$ is a factor of the latter two factors. We check that X^2+1 has no root in $\mathbb{F}_3$. Since this is a quadratic polynomial, this means that X^2+1 is irreducible over $\mathbb{F}_3$. The polynomial X^4+1 splits into two quadratic factors over $\mathbb{F}_3$. We can check that these are X^2+X-1 and X^2-X-1:

$$(X^2-1+X)(X^2-1-X) = (X^2-1)^2-X^2 = X^4-2X^2+1-X^2 = X^4-1 \text{ in } \mathbb{F}_3[X]$$

We note that

$$X^{27} - X = X(X^{26}-1) = X(X^2-1)\Phi_{13}(X)\Phi_{26}(X)$$

so that any irreducible polynomial of degree 3 over $\mathbb{F}_3$ occurs in the latter two factors, both of which are degree 12. In fact, $\Phi_{2k}(X) = \Phi_k(-X)$ for any *odd* number k. Hence $\Phi_{26}(X) = \Phi_{13}(-X)$. So we only need to find the factors of $\Phi_{13}(X)$. One expects 4 factors of degree 3 each.

A cubic polynomial is irreducible if and only if it has no root. We also note that $\alpha^3 = \alpha$ for an element of $\alpha \in \mathbb{F}_3$. As seen above, the quadratic polynomials $X^2 + X - 1$ and $-X^2 + X + 1$ have no roots in $\mathbb{F}_3$. It follows that the cubic polynomials $X^3 + X^2 - 1$ and $X^3 - X^2 + 1$ are irreducible over $\mathbb{F}_3$. Similarly, we note that $X^3 - X + 1$ and $X^3 - X - 1$ are irreducible over $\mathbb{F}_3$. Finding the remaining two irreducible polynomials is left as an exercise!

General p, degree p. Note that $P_k(X) = X^p - X - k$ has no roots in $\mathbb{F}_p$ for a a non-zero element of $\mathbb{F}_p$. One notes that $P_k(kX) = kP_1(X)$ for $k \in \mathbb{F}_p$. Hence, it is enough to show that $P_1(X)$ is irreducible.

Note that if E is a field in which $P_1(X)$ splits and $\alpha \in E$ is a root, then $\alpha + k$ is a root for every $k \in \mathbb{F}_p$. In fact, we have $\phi_p(\alpha) = \alpha + 1$. Thus, if $\phi_p^k(\alpha) = \alpha$, then k is a multiple of p. This shows that the field which contains the root of $P_1(X)$ is $\mathbb{F}_{p^p}$. Thus, $P_1(X)$ is irreducible.

General p, any degree. There are two well-known algorithms for factoring polynomials over finite fields:

- Berlekamp's Algorithm
- The Cantor-Zassenhaus Algorithm

The second one is used in practice on computers as it is faster.